

Sham Satish Thakare

Independent Computer Science Researcher

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RESEARCH STATEMENT & INTERESTS

I conduct foundational and systems research on **Trustworthy, Adaptive, and Verifiable Autonomous Systems**. My work focuses on four primary vectors:

Foundation Model Plasticity & Calibrated Reasoning: Representation geometry diagnostics (α_{SVD} , $\bar{H}$, σ_g^2) for predicting reinforcement learning reward plasticity in intermediate checkpoints, token entropy length confounding, and RLVR self-consistency calibration.

Confidential Computing & Hardware Security: Zero-knowledge remote attestation membership proofs and access-frequency-weighted adaptive Oblivious RAM (ORAM) tree rebalancing algorithms for hardware enclaves (SGX/TDX).

Learning-Augmented Distributed Consensus: Failure-domain aware dynamic quorum adaptation (AdaptiveReplica) over Raft joint-consensus transitions with uncertainty-aware trust gates (T_3).

Causal Observability & Verifiable AI Systems: Topological causal walks over OpenTelemetry microservice dependency graphs (TraceMind) and zero-knowledge authorization-path provenance graph proofs (EnclaveShield).

EDUCATION

Anantrao Pawar College of Engineering & Research (APCOER), Pune, India

Bachelor of Technology in Artificial Intelligence and Data Science | Jan 2020 – Jun 2024

- **Cumulative GPA:** 8.70 / 10.00

- **Core Coursework:** Engineering Mathematics, Scientific Computing, Operating Systems, Design and Analysis of Algorithms, Object-Oriented Programming (C++), Python Programming, Data Structures, Computer Networks, Database Management Systems, Artificial Intelligence, Machine Learning, Computational Modeling, Signal Processing, Linear Algebra, Probability and Statistics.

RESEARCH EXPERIENCE & MANUSCRIPTS

Active Submitted Manuscripts

When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LM RL

Author: Sham Satish Thakare

Status: **Submitted / Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI)** (Submitted Aug 2026)

Repository: [ear_grpo_reasoning](https://github.com/shamddd/ear_grpo_reasoning)

Key Finding: Exposes length confounding in token predictive entropy ($R = +0.486$). Sample-level consensus weighting (CA-GRPO) yields 0.00% Pass@1 gain over standard outcome-supervised GRPO across 3 matched seeds.

recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

Author: Sham Satish Thakare

Status: **Submitted / Under Review at 11th IEEE Special Session on Machine Learning on Big Data (MLBD 2026) / IEEE**

BigData 2026 (Submission ID: `BigD497`, Submitted Aug 17, 2026)

Repository: [recovery_eval](https://github.com/shamddd/recovery_eval)

Key Finding: Matched state-contrast protocol ($D_{\text{recovery}} = -0.1100$, 95% CI $[-0.240, +0.030]$) across 400 continuations of Qwen2.5-Math 1.5B.

StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning

Author: Sham Satish Thakare

Status: **Submitted / Under Review at Elsevier Artificial Intelligence (AIJ)** (Manuscript: `ARTINT-D-26-01491`, Submitted 2026)

Repository: [stateshift](https://github.com/shamddd/stateshift)

Key Finding: Formulates verifier-defined continuation state matching on neural reasoning rollouts under structural covariate controls.

Preprints & Working Papers

Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?

Author: Sham Satish Thakare

Status: **Working Paper** (Target under consideration: *Transactions on Machine Learning Research - TMLR*) (2026)

Repository: `submission/tmlr` / `submission\_package`

Key Finding: Formalizes deployment cost frontiers ($C_{\text{total}} = C_{\text{train}} + Q \cdot C_{\text{inference}}$). Demonstrates OOD length extrapolation accelerates RLVR amortization crossover ($R_f \approx 0.0618 \parallel 1.0$).

Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints: A Cross-Architecture Diagnostic Study

Author: Sham Satish Thakare

Status: **Working Paper** (Target under consideration: *Journal of Machine Learning Research - JMLR*) (2026)

Repository: [adaptive-rl-forge](https://github.com/shamddd/adaptive-rl-forge)

Key Finding: Formulated diagnostic probing vectors $\phi(C_k)$ predicting downstream GRPO reward gain β_{RL} with $R^2 = 0.91$ ($p = 0.0004$).

EnclaveShield: Zero-Knowledge Memory Attestation and Side-Channel Mitigation for Hardware Enclaves

Author: Sham Satish Thakare

Status: **Working Paper** (Target under consideration: *IEEE Transactions on Dependable and Secure Computing - TDSC*) (2026)

Repository: [enclaveshield](https://github.com/shamddd/enclaveshield)

Key Finding: Engineered ZK quote attestation membership proofs and access-frequency-weighted adaptive Path ORAM tree rebalancing (1.47 ms latency, $H(A) = 0.82 \text{ ppm}$ 0.02).

AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus

Author: Sham Satish Thakare

Status: **Working Paper** (Target under consideration: *IEEE Transactions on Parallel and Distributed Systems - TPDS*) (2026)

Repository: [quorumshift](https://github.com/shamddd/quorumshift)

Key Finding: Formulated dynamic vote-weight adaptation over Raft joint-consensus transitions with trust gates (T_3), reducing p99 write latency to 13.50 ms (88.8% reduction vs static 120.48 ms) with zero stale reads.

TraceMind: Graph-Constrained Causal Reasoning for Root-Cause Localization in Microservice Systems

Author: Sham Satish Thakare

Status: **Working Paper** (Target under consideration: *IEEE Transactions on Cloud Computing - TCC*) (2026)

Repository: [tracemind](https://github.com/shamddd/tracemind)

Key Finding: Constructed topological causal walks over OpenTelemetry Service Dependency Graphs, achieving 100.0% Top-1 RCA accuracy (MRR = 1.00) across 24 fault scenarios.

PROFESSIONAL & ACADEMIC EXPERIENCE

Spectra Corporate Service, Pune, India

Data Engineer / Machine Learning Specialist | Jun 2024 – Present

- Developed and maintained scalable data processing pipelines and ML-support workflows using Python, SQL, and GCP/AWS cloud services.
- Built and debugged automated CI/CD deployment pipelines, containerized infrastructure, and monitoring dashboards using Docker, Kubernetes, and Prometheus/Grafana.
- Engineered backend data integration services and performance-monitoring workflows for production model pipelines.

APCOER Pune, Pune, India

Research Intern | Jun 2023 – Dec 2024

- Assisted faculty in experimental design, statistical analysis, and open-source software reproducibility protocols.
 - Provided teaching assistance and course material support for undergraduate courses including **Intro to Algorithms** and **Foundations of ML** (Aug 2024 – Dec 2024).
 - Contributed to course content preparation and assignment design for **Fundamentals of Computation**, **Introduction to Algorithms**, and **Theory of Computation** (Aug 2022 – May 2023).
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TECHNICAL SKILLS

- **Research & Machine Learning:** PyTorch, GRPO / RLVR Probes, Transformers, HuggingFace, NumPy, SciPy, Scikit-Learn
- **Systems Research Expertise** *(Demonstrated in Open-Source Research Artifacts)*: Raft Consensus ('quorumshift'), Path ORAM & ZK Attestation ('enclaveshield'), OpenTelemetry Causal Graphs ('tracemind')
- **Professional Data & Cloud Engineering:** Python, SQL, GCP, AWS, Docker, Kubernetes, Prometheus, Grafana
- **Programming Languages:** Python, C++, C, SQL, Shell (Bash), LaTeX